

ZARA



The Zara Pricing Effect: How Price and Promotion Drive Sales Performance



Industry

Fast fashion >> highly competitive Zara needs fast decisions

- The fashion retail industry is highly competitive and fast-paced
- Companies like **Zara** must quickly respond to changing consumer demand
- Pricing, promotions, and product placement directly impact sales performance
- Data-driven insights are essential for optimizing retail strategy



Business Problem

ZARA



- Zara operates in a fast-fashion market where decisions must be made quickly
- It is unclear which factors most strongly drive sales performance
- Ineffective pricing, promotions, or placement can reduce profitability

Goal >> Identify key drivers of sales to improve retail decision - making

Project Objective

Understanding what drives Zara's sales performance

- Analyze Zara's sales data using visual analytics
- Identify factors influencing product performance
- Evaluate the impact of pricing, promotions, and product positioning
- Provide insights to improve retail strategy



Research Questions

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Which factors (price, discount, product category, store location, promotion type) most strongly influence Zara's sales performance?

How do pricing and promotional strategies impact total sales and revenue?

Do discounted products sell significantly more units than non-discounted items?

Which product categories respond most positively to promotions?

Literature Review

Economic Implications



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The Economic Implications of Fast Fashion

- High-volume, trend driven consumption
- Lower & mid-priced items resulting in higher sales volume

Brand Competition in Fashion E-commerce

- Consumers respond to price changes
- Strong brand loyalty limits switching
- Zara leverages pricing + brand



Data & Analytical Approach

Methodology

- Data cleaning and preparation (handling missing values, standardizing variables)
- Creation of key variables (Price Category, Revenue)
- Exploratory Data Analysis (EDA) using R Studio
- Application of visual analytics techniques (bar charts, line charts, scatterplots)
- Comparative and trend analysis across pricing, categories, and promotions





Data Overview

Source: **Kaggle Zara dataset** 700 product observations used to analyze how different factors influence sales

Dependent Variables

Independent Variables

Sales Volume
Total Amount (Revenue)

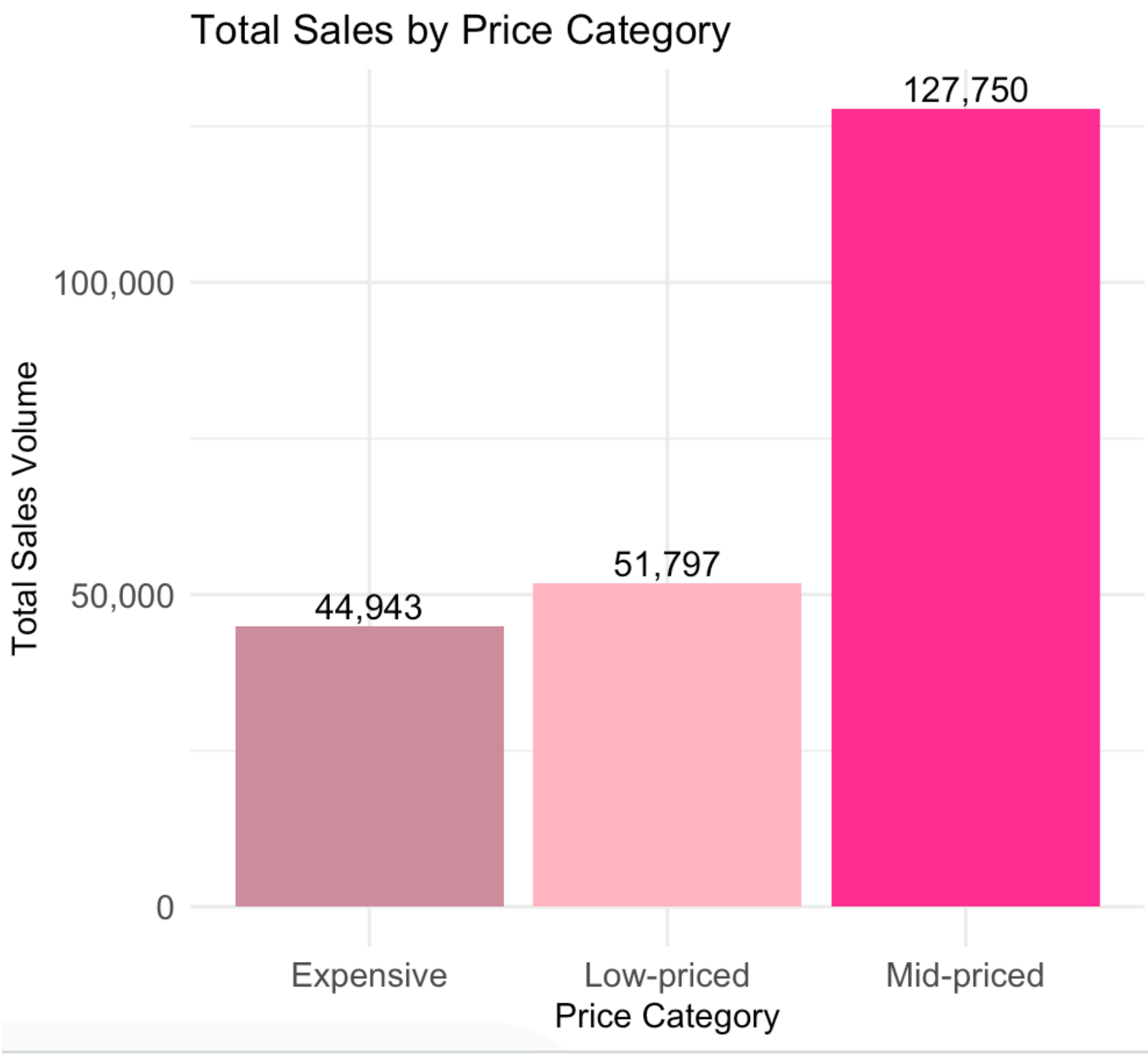
Price
Price Category
Promotion
Product Category
Section
Product Position
Seasonal

Data

zara																
Product	Product Post	Promo	Product Category	Season	Sales Volume	brand	sku	name	description	Price	price	Total Amount	scraped_at	item	Sex	Price Category
951174	Aisle	No	Clothing	No	523	Zara	112595203-659-2	OVERSIZED HOODIE	Oversized Hoodie made with premium fabric and modern tailoring.	70.43	69.95	36593.85	2024-02-20T00:00:00	tops	MAN	Mid-priced
626511	End-cap	Yes	Clothing	Yes	348	Zara	909557208-699-1	PLEATED SKIRT	Pleated Skirt made with premium fabric and modern tailoring.	81.65	81.95	28518.6	2024-02-20T00:01:00	coats	WOMAN	Mid-priced
293253	End-cap	Yes	Clothing	Yes	92	Zara	496257552-888-3	TAILORED WOOL COAT	Tailored Wool Coat made with premium fabric and modern tailoring.	244.2	243.95	22443.4	2024-02-20T00:02:00	dresses	WOMAN	Expensive
620974	End-cap	Yes	Clothing	Yes	219	Zara	785105029-554-1	RELAXED FIT BLAZER	Relaxed Fit Blazer made with premium fabric and modern tailoring.	242.46	241.95	52987.05	2024-02-20T00:03:00	coats	WOMAN	Expensive
753289	End-cap	No	Clothing	Yes	72	Zara	610660545-530-2	RELAXED FIT BLAZER	Relaxed Fit Blazer made with premium fabric and modern tailoring.	243.17	242.95	17492.4	2024-02-20T00:04:00	blazers	WOMAN	Expensive
779994	Aisle	No	Clothing	No	275	Zara	328550722-423-5	PLEATED SKIRT	Pleated Skirt made with premium fabric and modern tailoring.	151.05	150.95	41511.25	2024-02-20T00:05:00	pants	WOMAN	Mid-priced
356910	End-cap	No	Clothing	No	140	Zara	572070500-584-3	OVERSIZED HOODIE	Oversized Hoodie made with premium fabric and modern tailoring.	240.55	240.95	33733	2024-02-20T00:06:00	dresses	MAN	Expensive
826254	Aisle	No	Clothing	No	652	Zara	879010707-167-1	PLEATED SKIRT	Pleated Skirt made with premium fabric and modern tailoring.	74.97	74.95	48867.4	2024-02-20T00:07:00	pants	MAN	Mid-priced
428833	Aisle	No	Clothing	No	606	Zara	472245143-972-5	TAILORED WOOL COAT	Tailored Wool Coat made with premium fabric and modern tailoring.	129.15	128.95	78143.7	2024-02-20T00:08:00	pants	MAN	Mid-priced
485592	End-cap	Yes	Clothing	Yes	271	Zara	900635895-921-4	TAILORED WOOL COAT	Tailored Wool Coat made with premium fabric and modern tailoring.	174.17	173.95	47140.45	2024-02-20T00:09:00	t-shirts	MAN	Expensive
477232	End-cap	Yes	Clothing	Yes	166	Zara	319859325-722-4	BASIC COTTON TEE	Basic Cotton Tee made with premium fabric and modern tailoring.	243.87	243.95	40495.7	2024-02-20T00:10:00	blazers	MAN	Expensive
567708	Aisle	Yes	Clothing	No	248	Zara	957537886-356-1	SATIN MIDI DRESS	Satin Midi Dress made with premium fabric and modern tailoring.	119.66	119.95	29747.6	2024-02-20T00:11:00	blazers	MAN	Mid-priced
598191	End-cap	No	Clothing	No	161	Zara	326012825-158-2	SATIN MIDI DRESS	Satin Midi Dress made with premium fabric and modern tailoring.	45.11	44.95	7236.95	2024-02-20T00:12:00	pants	MAN	Low-priced
872816	End-cap	No	Clothing	Yes	517	Zara	477241354-962-1	TAILORED WOOL COAT	Tailored Wool Coat made with premium fabric and modern tailoring.	110.92	110.95	57361.15	2024-02-20T00:13:00	tops	MAN	Mid-priced
264240	Aisle	No	Clothing	No	650	Zara	898382888-460-3	TAILORED WOOL COAT	Tailored Wool Coat made with premium fabric and modern tailoring.	113.27	112.95	73417.5	2024-02-20T00:14:00	dresses	MAN	Mid-priced
299982	End-cap	Yes	Clothing	No	578	Zara	696668080-619-3	BASIC COTTON TEE	Basic Cotton Tee made with premium fabric and modern tailoring.	79.62	79.95	46211.1	2024-02-20T00:15:00	coats	MAN	Mid-priced
607401	End-cap	Yes	Clothing	No	205	Zara	494342825-270-3	WIDE LEG PANTS	Wide Leg Pants made with premium fabric and modern tailoring.	188.51	188.95	38734.75	2024-02-20T00:16:00	pants	MAN	Expensive
754599	End-cap	No	Clothing	Yes	191	Zara	715488808-213-5	RELAXED FIT BLAZER	Relaxed Fit Blazer made with premium fabric and modern tailoring.	171	170.95	32651.45	2024-02-20T00:17:00	Pants	MAN	Expensive
835195	Front of Store	No	Clothing	No	488	Zara	645286103-617-4	PLEATED SKIRT	Pleated Skirt made with premium fabric and modern tailoring.	75.28	74.95	36575.6	2024-02-20T00:18:00	Pants	MAN	Mid-priced
525656	Aisle	No	Clothing	No	193	Zara	395565388-962-2	OVERSIZED HOODIE	Oversized Hoodie made with premium fabric and modern tailoring.	234.24	233.95	45152.35	2024-02-20T00:19:00	t-shirts	MAN	Expensive
704377	Aisle	No	Clothing	Yes	301	Zara	898236588-660-2	CROPPED JACKET	Cropped Jacket made with premium fabric and modern tailoring.	146.24	145.95	43930.95	2024-02-20T00:20:00	Pants	MAN	Mid-priced

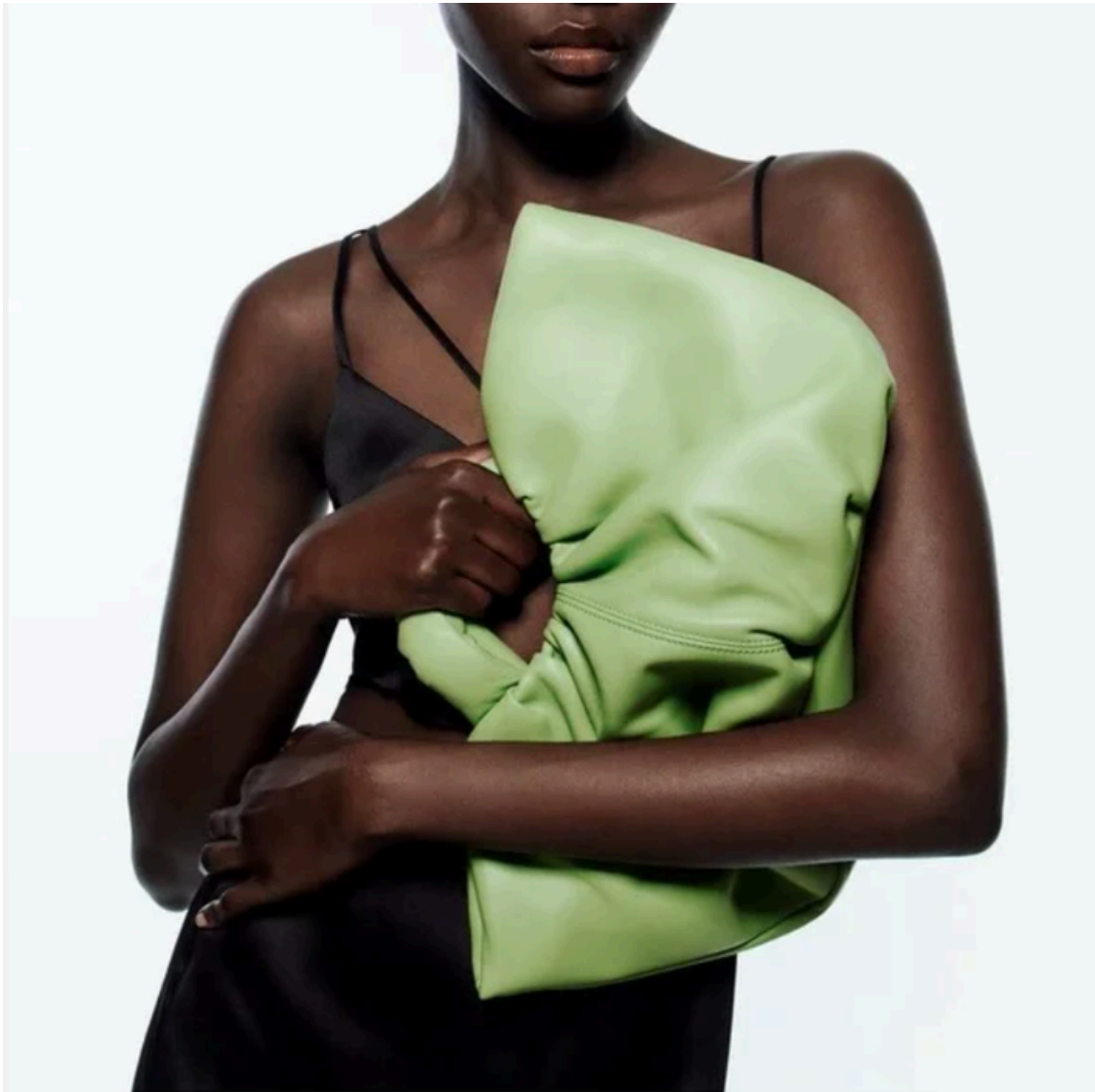


Price Analysis

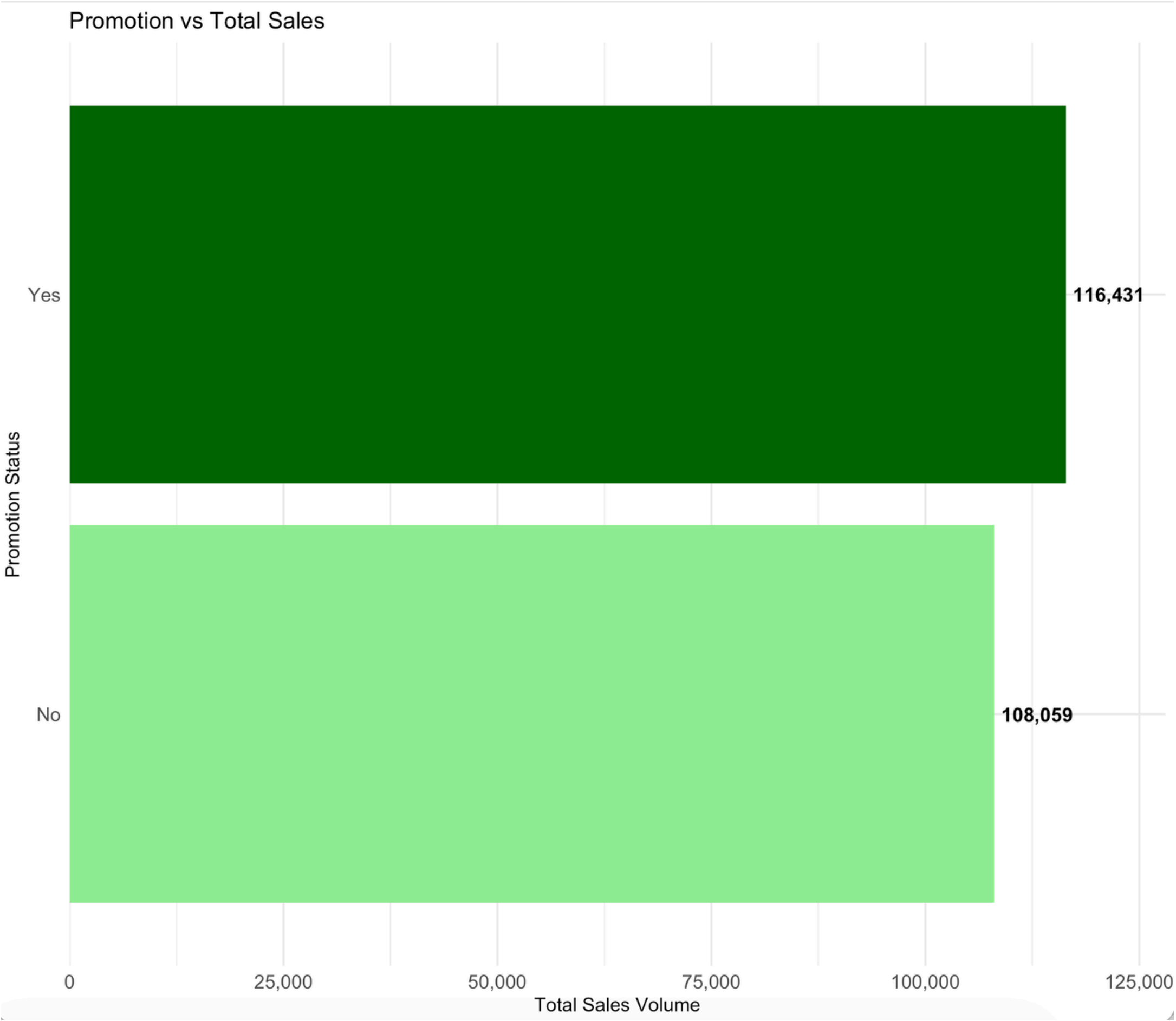


Mid-priced products generate the highest sales volume, significantly outperforming both low-priced and expensive categories. This indicates that customers prefer products that offer a balance between affordability and perceived value

Promotion vs Total Sales



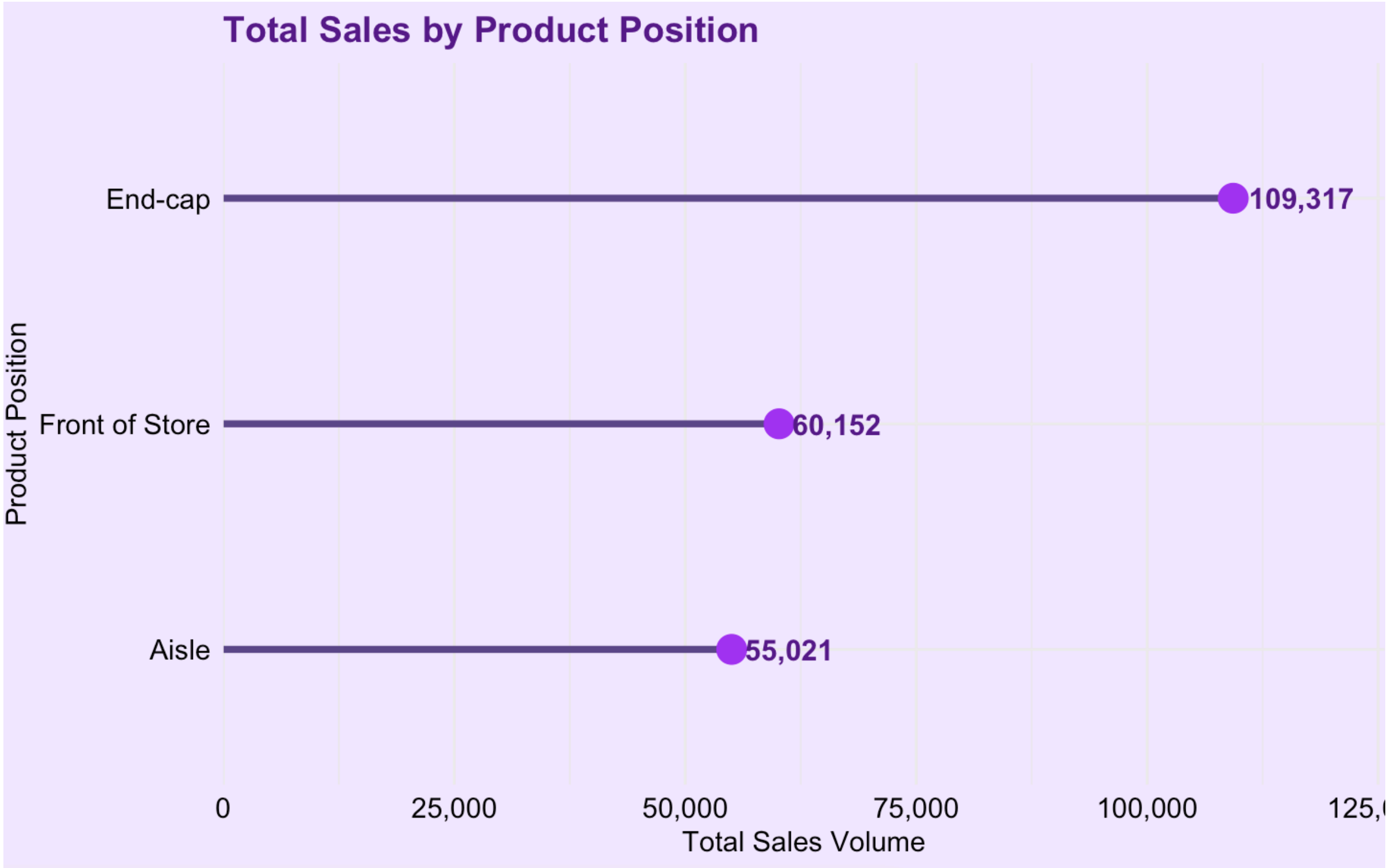
Promoted products generate higher sales volumes, suggesting that Zara should strategically use promotions to boost demand while ensuring they are targeted to maximize effectiveness and avoid excessive discounting.



Sales by Product Position



Total Sales by Product Position



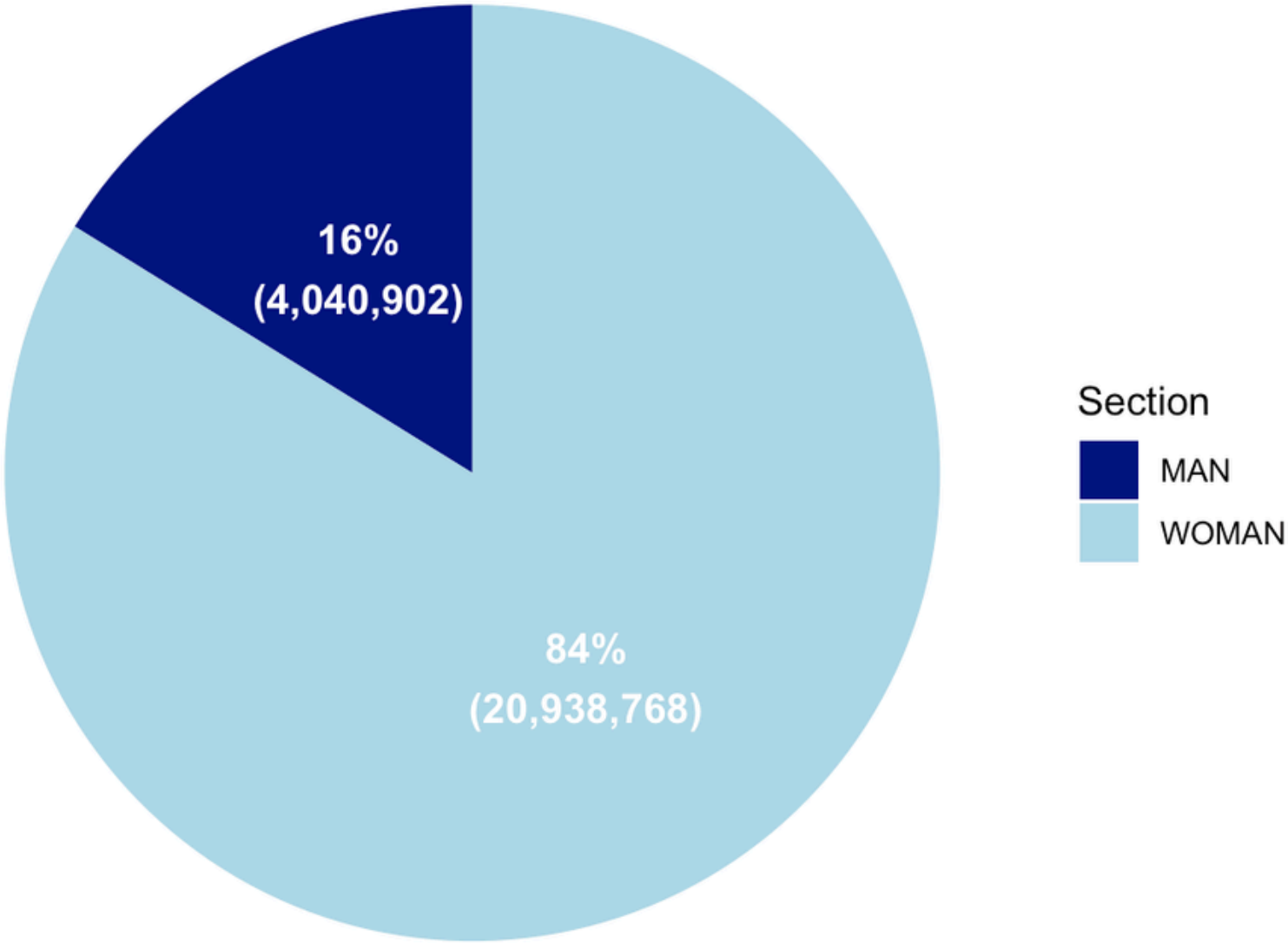
We used a **lollipop chart** to highlight differences in sales across product positions.

This chart shows that product placement plays a major role in sales performance. Items placed in more visible locations tend to sell more, highlighting the importance of effective merchandising strategies.



Section Comparison

Revenue Distribution by Section



Sales are primarily driven by the Women’s section, which accounts for the largest share of total sales. This highlights stronger demand in this segment and suggests potential to improve performance in the Men’s section

Product Category & Total Sales Volume

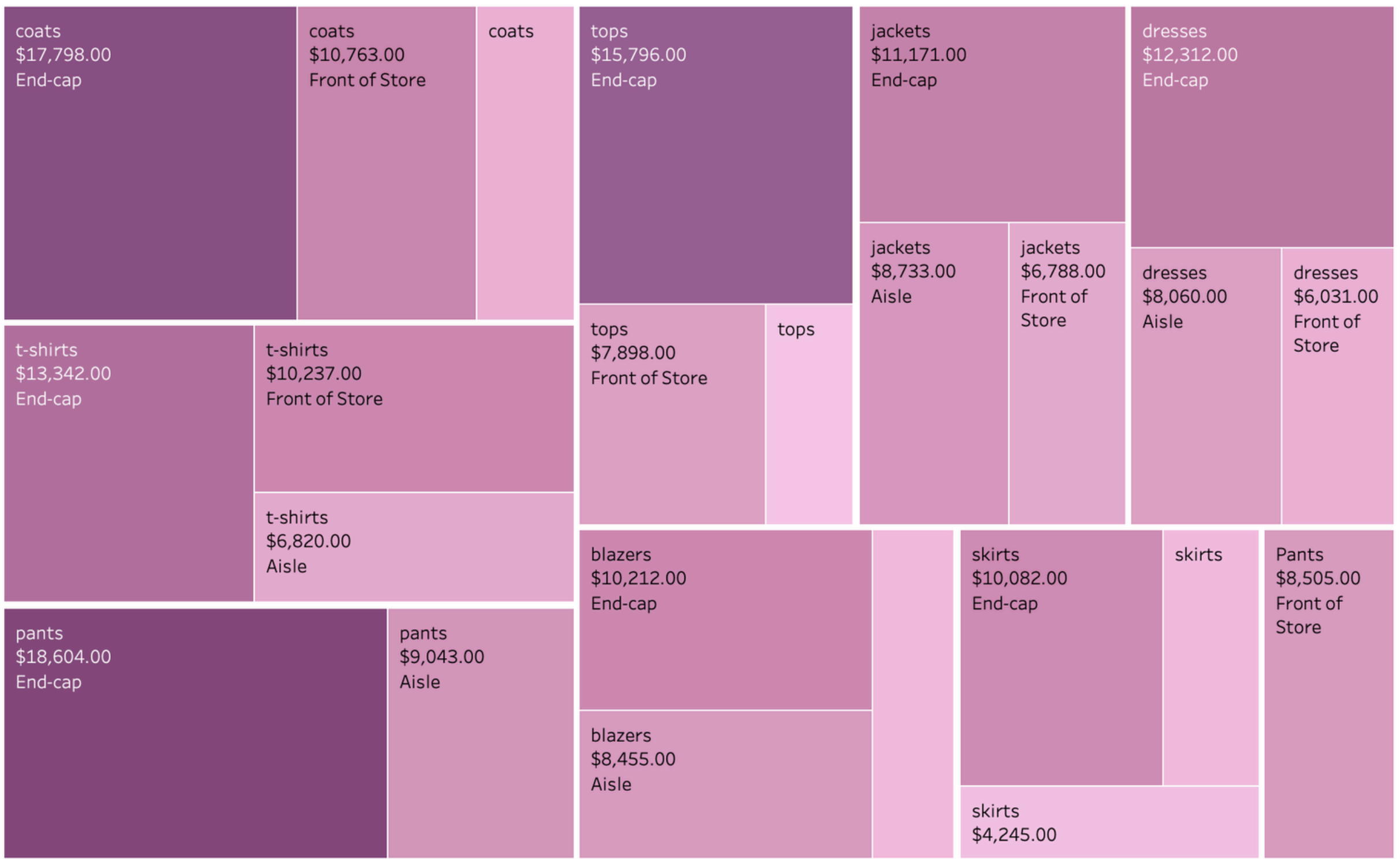


This treemap shows that **end-cap** placements consistently generate the highest sales across all product categories especially for

- pants, coats, and tops

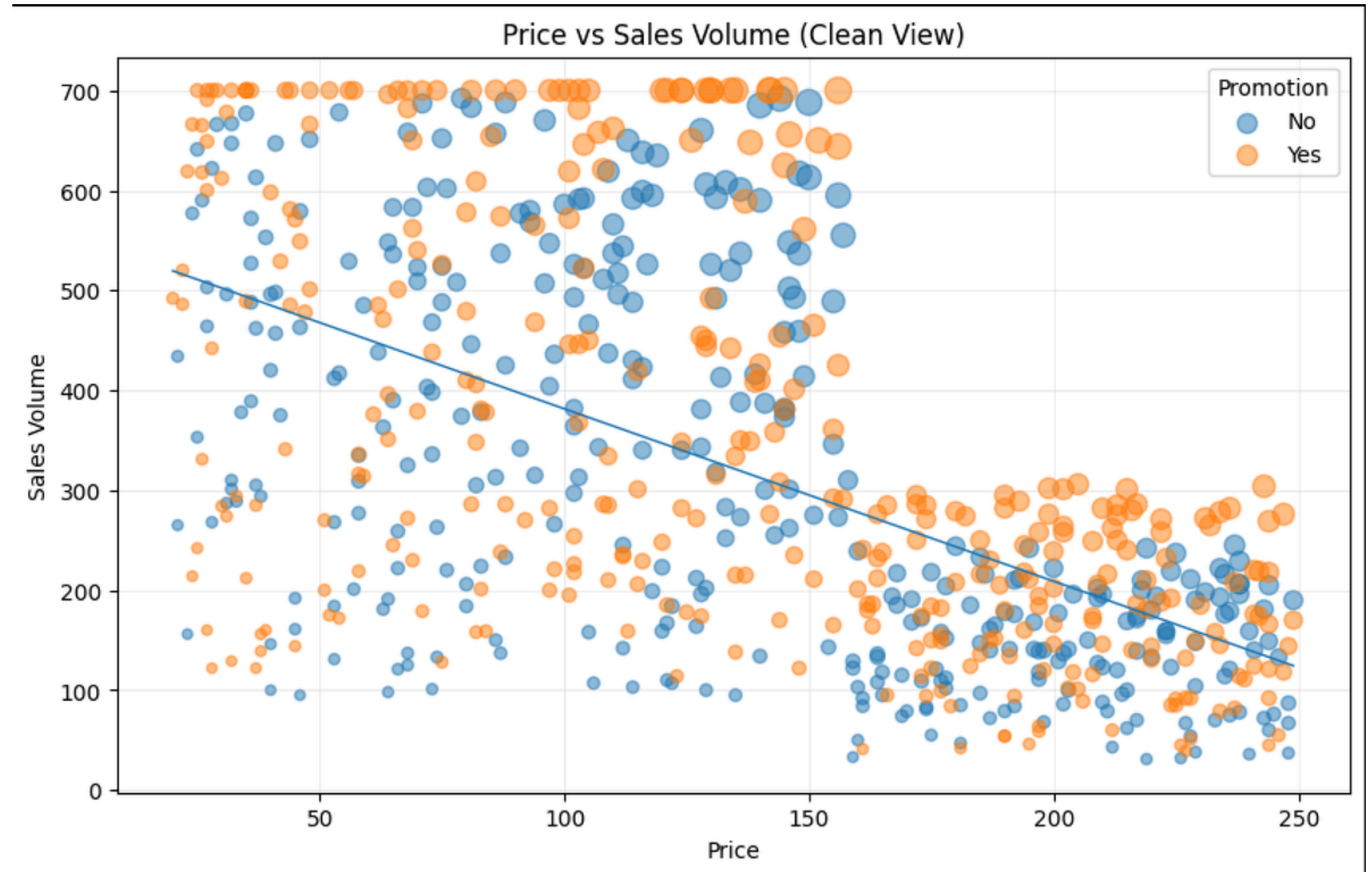
While aisle and front-of-store placements contribute less

Highlighting the strong impact of premium in-store positioning on revenue.



Price >> Volume Relationship

- \$150 Performance Cliff: Sales capacity drops from 700 units to <300 units once price exceeds \$150.
- Inverse Correlation: Steep negative slope confirms high price sensitivity across most categories.
- Promotion Density: Orange clusters sustain higher volume peaks in the \$100-\$150 mid-range.
- Premium products exhibit lower but more stable demand, indicating consistent performance in the luxury segment.



Key Insights

- Sales performance is influenced by multiple factors.
- Pricing, product category, and product placement emerge as the strongest drivers of sales performance.
- Mid-priced products generate the highest sales, indicating a clear balance between affordability and perceived value.
- Promotions and product visibility further enhance demand, highlighting the importance of strategic merchandising and targeted marketing.
- Data-driven insights can significantly improve retail decision-making and overall business performance.



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Future Direction

- Move to real-time, not retrospective analytics
- Predict demand, don't react to it → Leverage predictive modeling (ML)
- Segment customers, not averages → target high-value shoppers
- Localize strategy by region → align pricing & inventory with demand
- Integrate omnichannel data → unify online & in-store insights
- Maximize impact, protect margins
- Build interactive dashboards → enable continuous decision-making





Limitations

- The dataset used was relatively limited, short time frame with no seasonal long term data
- Customer-level data not included:
 - Demographic
 - Purchasing behaviour
- External Variables
 - competitor price drops, macroeconomic shifts, or viral social media trends that trigger overnight stock-outs
- Lack of Qualitative Context:
 - data cannot measure the emotional brand connection or "perceived quality" that often justifies a higher price point.





Interpretation of findings:

- Consumers are price-sensitive but stay brand-loyal
- Zara balances affordability with premium perception
- Demand is trend-driven and seasonal
- Switching behavior is limited due to brand strength



Practical implications:

- *Use dynamic pricing for better sales performance*
- *Invest in brand loyalty to reduce switching*
- *Use dashboards for inventory + trend forecasting*
- *Align marketing with real-time consumer demand*





Conclusion

- Sales are driven, not random → Pricing & promotions are key levers
- Not all discounts work → Targeted promotions outperform blanket discounts
- Invest in high-performing categories.
- Mid-price = sweet spot → Maximizes both volume & revenue
- Data enables smarter, faster decisions



Thank You



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Any

Questions?

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